Boundary protection without score coverage

Teacher scores and cheap predictions $k = 2, \quad \sigma_i = 1$

True winners

d_1	d_2	d_3	d_4
$s = 9$	$s = 8$	$s = 7$	$s = 1$
$\mu = 8$	$\mu = 6$	$\mu = 6$	$\mu = -100$

$s_{(k+1)} = 7$

Required threshold

Rank-sufficient — Reach the boundary — $A_{2,1} = 1$

Winner coverage — Cover the winners — $B_{\text{top}} = 2$

All-score coverage — Pay for the irrelevant tail — $B_{\text{all}} = 101$

Cheap-score order $d_1 \to d_2 \to d_3 \to d_4$

$\lambda = 0.5$ is safe on this trajectory

Changed tie-break $d_1 \to d_3 \to d_2 \to d_4$

The winner d_2 is scored later

Same unsafe scored set $V = \{d_1, d_3\}$ Winner d_2 remains unscored $\tau(V) = 7$

Safe on one order $\lambda = 0.5$

$$\tau(V) = 7 > \max(6.5, -99.5)$$

Wrong stop
Returns d_3 for d_2

Rank-sufficient threshold $\lambda = 1$

$$\tau(V) = 7 = \max(7, -99)$$

Stop blocked at equality
The stopping inequality is strict.